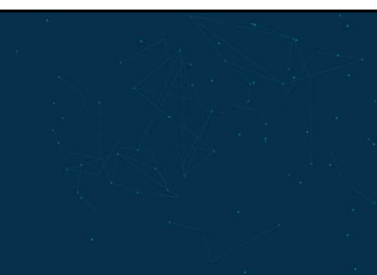


Network-Based Attacks

CHAPTER 3



3.01

Episode **Network-Based Exploits**
title:

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**

NAME RESOLUTION EXPLOITS

- **NETBIOS (NBNS) name service**
 - Part of NetBIOS-over-TCP
 - Similar functionality to DNS
- **Link-local Multicast Name Resolution (LLMNR)**
 - Protocol based on DNS packet format
 - Allows IPv4 and IPv6 name resolution on the same local link
- **DNS and ARP poisoning**

NETWORK EXPLOITS

- **Server Message Block (SMB) exploits**

- Protocol used in Windows to provide file and printer access, and remote service access
- TCP ports 139 and 445
- Some ransomware (EternalBlue, WannaCry) use SMB to propagate

- **Simple Network Management Protocol (SNMP) exploits**

- Query and manage IP devices
- Multiple versions - SNMPv1 is not secure

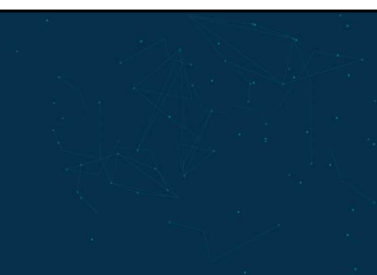
EVEN MORE NETWORK EXPLOITS

- **Simple Mail Transport Protocol (SMTP) exploits**

- Standard protocol for transmitting email
- Open relay, local relay, phishing, spam, etc.

- **File Transfer Protocol (FTP) exploits**

- Overall insecure protocol for transferring files
- No encryption for transfers and credentials
- Easy for attackers to use for data exfiltration if FTP is available



3.02

Episode **FTP Exploit Demo**
title:

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**

FTP Exploit Demo

- FTP demo exploit

3.03

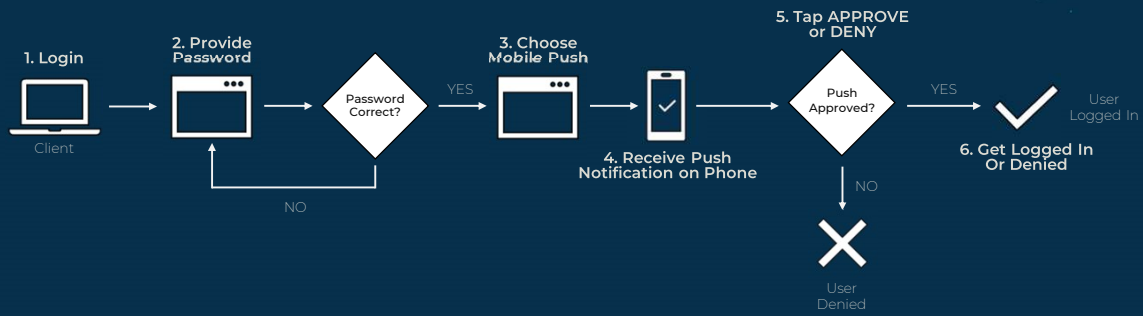
Episode **Authentication Attacks**
title:

Objective: **4.3.1.1, 4.3.1.9, 4.3.1.12, 4.3.1.13**

MFA FATIGUE ATTACKS

- Human side exploitation
- Push-based systems
- Repeated login attempts
- Lack of training

MULTI-FACTOR AUTHENTICATION PUSH-BASED SYSTEM



MASK ATTACKS

- **Optimized brute force**
- **Defined patterns**
- **Tools:**
 - Hash Cat
 - John the Ripper

```
$ hashcat -a 3 -m 0 hashes.txt ?u?l?l?d?d?d
```

```
hashcat (vX.X.X) starting...
```

```
Progress.....: 00000123/4569760000 (0.00%)
```

```
Candidates....: Abcd1234 -> Zzzz9999
```

```
Trying.....: John1234
```

```
Trying.....: Mark5678
```

```
Trying.....: Test0001
```

```
Password found: John1234
```

OIDC VULNERABILITIES

- Open ID Connect (OIDC)
- Federated identity protocol
- Token substitution
- Open redirect vulnerabilities

SAML ATTACKS

- XML-based protocol
- Signature wrapping
- Assertion spoofing
- High impact target

3.04

Episode title: **Man-in-the Middle Exploits**

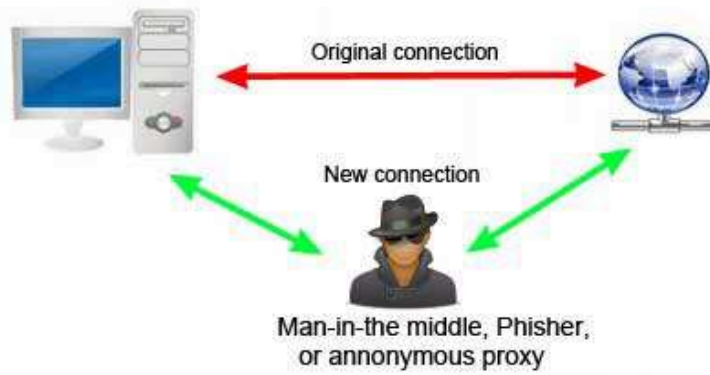
Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**
3.2 Given a scenario, research attack vectors and perform wireless attacks

ADDITIONAL NETWORK EXPLOITS

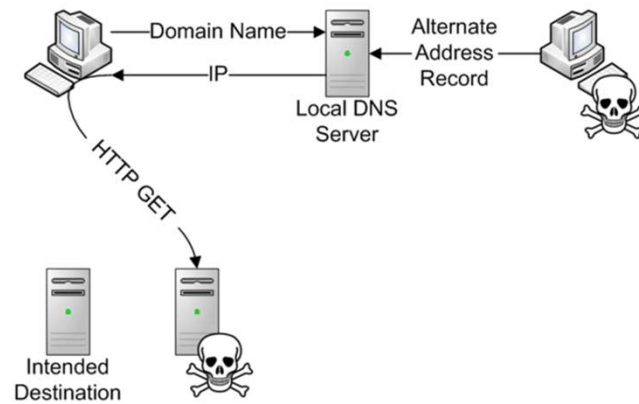
- **Man-in-the-middle**

- Family of attacks where the attacker intercepts messages between a sender and receiver
- Attack may modify, regenerate, or forward intercepted messages

Man-in-the-middle attack



DNS Cache Poisoning



MAN-IN-THE-MIDDLE EXPLOITS

- **ARP spoofing**

- Similar to DNS poisoning, but with local MAC addresses

- **Pass the hash**

- Attacker intercepts an NTLM hash (user credential) and reuses it to appear as an authenticated user to Windows

MAN-IN-THE-MIDDLE EXPLOITS

- **Replay**
- **Relay**
- **Secure Sockets Layer (SSL) stripping**
- **Downgrade**

MAN-IN-THE-MIDDLE EXPLOITS

- Denial of Service (DoS) /stress test
- Network Access Control (NAC) bypass
- Virtual Local Area Network (VLAN) hopping

3.05

Episode title: **Labtainers Lab (TCP/IP Attacks)**

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**

3.06

Episode title: **Labtainers Lab (ARP Spoof Attack)**

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**

3.07

Episode title: **Labtainers Lab (Local DNS Attacks)**

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**

3.08

Episode title: **Labtainers Lab (MACs and Hash Functions)**

Objective: **3.1 Given a scenario, research attack vectors and perform network attacks**